INTERNSHIP PROJECT REPORT

Kali Linux Lab Environment

System, Storage and Network Verification

Cyber Security practice machine running directly on a laptop

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Submitted to: _____

Date: 30 October 2026

1. Project Summary

In this project I checked my Kali Linux machine from top to bottom: the laptop it runs on, the memory, the storage, the operating system and the network. I used simple terminal commands and took a screenshot of every result. The goal was to be sure that this machine is ready to be used for cyber security and ethical hacking practice, and to write down exactly how it is set up.

Kali Linux is not running inside VirtualBox here. I boot it directly on the laptop and it runs from its own partitions on the internal SSD. This gives it full use of the real hardware, including the Wi-Fi card.

At a glance

Machine type	Laptop (physical machine, chassis: laptop)
Operating system	Kali GNU/Linux Rolling 2026.3
Kernel	Linux 7.1.5+kali-amd64 (x86-64)
User and hostname	prince @ kali
Memory (RAM)	7.4 GiB total, 2.9 GiB available
Storage	476.9 GB NVMe SSD, Kali root partition 96.4 GB
Network	Wi-Fi (wlan0), IP 192.168.1.18/24, gateway 192.168.1.1
Internet	Working - 4 of 4 pings to 8.8.8.8 received

2. Objectives

- Confirm that Kali Linux starts and works on the laptop hardware.
- Record the hardware details: machine type, memory and storage.
- Record the software details: user, operating system version and kernel.
- Record the network details: interface, IP address and gateway.
- Test the connection to a local address and to the Internet.
- Compare the result with the reference lab setup (WK1-PM1) and note the differences.

3. Method and Commands Used

I started Kali Linux from the boot menu of the laptop, logged in as **prince** and opened the Konsole terminal. I ran each command below and took a screenshot with the command and its output visible. The screenshots were cropped to remove empty borders. For privacy, I covered a few identifiers (machine ID, boot ID, laptop brand and model, MAC address and public IPv6 address) with black boxes.

Command	What it shows
`hostnamectl`	Chassis type, operating system, kernel and firmware
`free -h`	Total, used and free memory and swap
`lsblk`	Disks and partitions, and where Kali is mounted
`whoami`	The user that is logged in
`uname -a`	Kernel version and system type
`cat /etc/os-release`	Name and version of the operating system
`ip addr show`	Network interfaces and their IP addresses
`ip route`	Routing table and default gateway
`ping -c 4 <address>`	Whether another address can be reached, with packet loss and reply time

4. Hardware Check

Step 1: Device and system details

```
hostnamectl
```

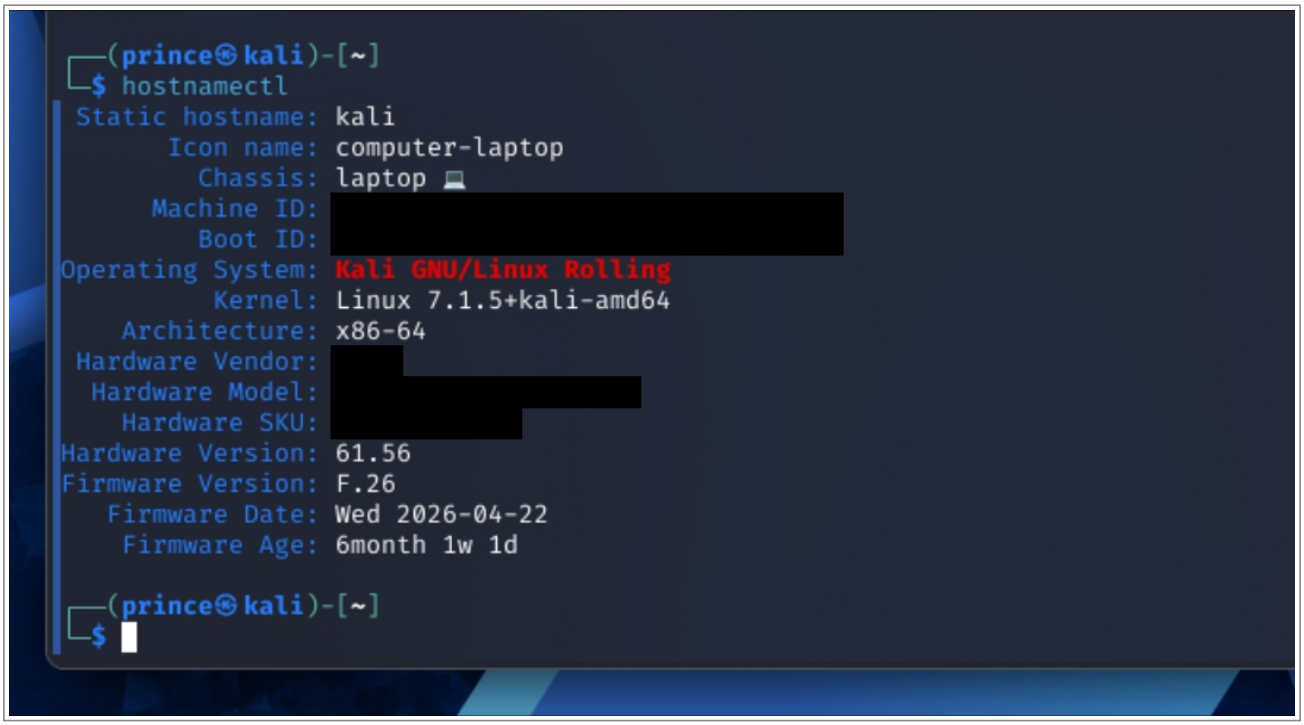


Figure 1: Output of hostnamectl

What I observed: The chassis is *laptop* and the icon name is *computer-laptop*. The firmware version is F.26, dated 22 April 2026. The operating system is Kali GNU/Linux Rolling with kernel 7.1.5+kali-amd64 on x86-64. Because the system reports a real laptop with real firmware, Kali is running on the physical machine. A VirtualBox virtual machine would normally show a virtual chassis and model here. I covered the machine ID, boot ID and the laptop brand and model with black boxes for privacy.

Step 2: Memory

```
free -h
```



Figure 2: Output of free -h

What I observed: Total memory is **7.4 GiB**. About 4.5 GiB was in use with the desktop and terminal open, 2.7 GiB was used as cache and **2.9 GiB was still available**. The swap space is 5.2 GiB and none of it was used (0 B), so the memory was enough at that time. The reference lab recommends 8 GB RAM or more, and this laptop is close to that.

Step 3: Storage and partitions

```
lsblk
```

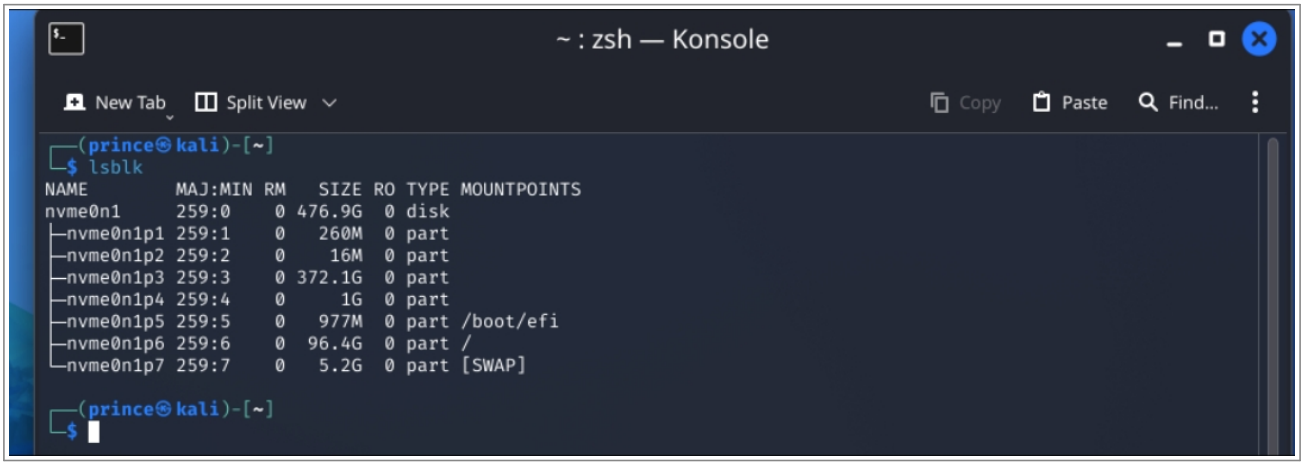


Figure 3: Output of lsblk

What I observed: The internal disk is **nvme0n1**, a 476.9 GB NVMe SSD, split into seven partitions. Kali's root file system `/` is on **nvme0n1p6 (96.4 GB)**, its boot files `/boot/efi` are on nvme0n1p5 (977 MB) and the swap is on nvme0n1p7 (5.2 GB). The other partitions (p1 to p4, including the 372.1 GB one) are not mounted in Kali, so Kali is not using them. The disk is larger than the 256 GB SSD recommended in the reference.

Partition summary (from Figure 3)

Partition	Size	Mount point	Used for
nvme0n1p1	260 MB	-	Not mounted in Kali
nvme0n1p2	16 MB	-	Not mounted in Kali
nvme0n1p3	372.1 GB	-	Not mounted in Kali
nvme0n1p4	1 GB	-	Not mounted in Kali
nvme0n1p5	977 MB	`/boot/efi`	Boot files
nvme0n1p6	96.4 GB	`/`	Kali Linux system and files
nvme0n1p7	5.2 GB	[SWAP]	Swap space

5. Software Check

Step 4: Logged in user

```
whoami
```

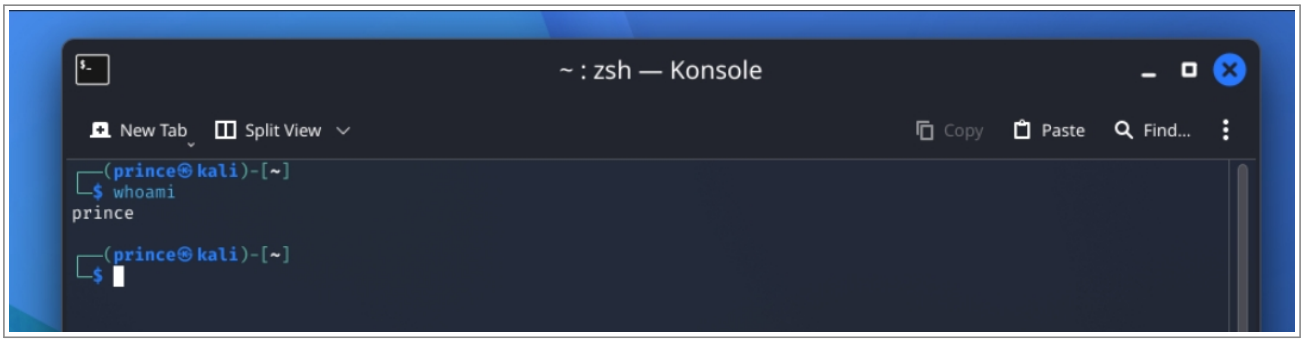


Figure 4: Output of whoami

What I observed: The output is **prince**, and the prompt shows `(prince@kali)-[~]`. So the user name is prince and the hostname is kali.

Step 5: Kernel and Kali version

```
uname -a
cat /etc/os-release
```

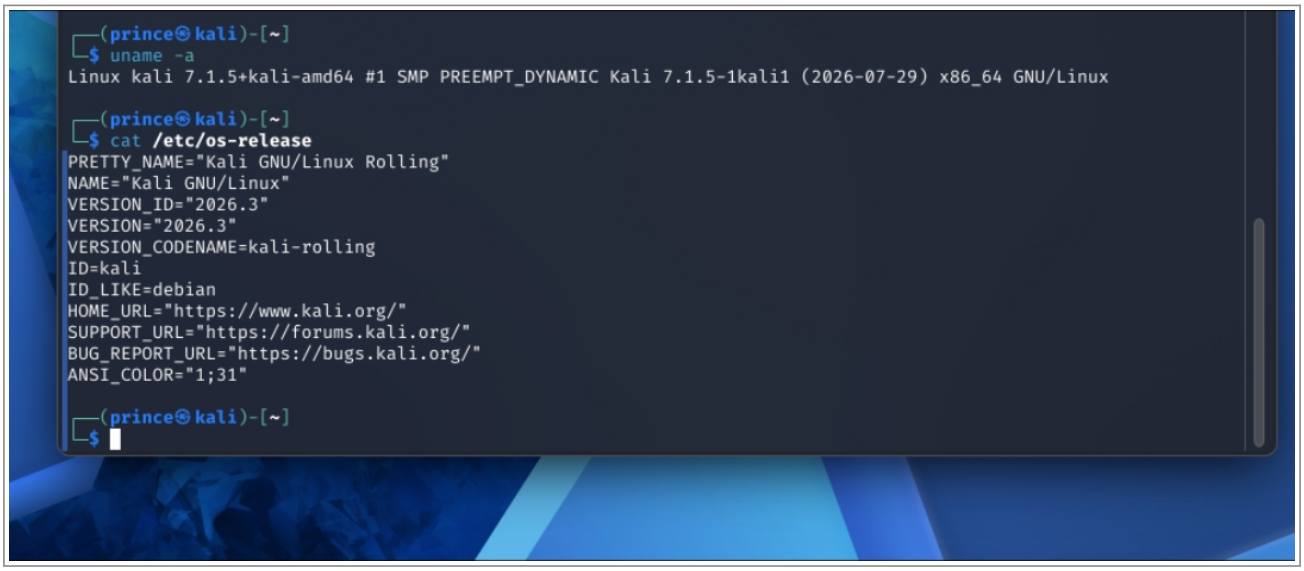


Figure 5: Output of uname -a and cat /etc/os-release

What I observed: The kernel is **Linux 7.1.5** (kali-amd64), built on 2026-07-29, 64-bit. The system is **Kali GNU/Linux Rolling, version 2026.3**, based on Debian.

6. Network Check

Step 6: IP address

```
ip addr show
```

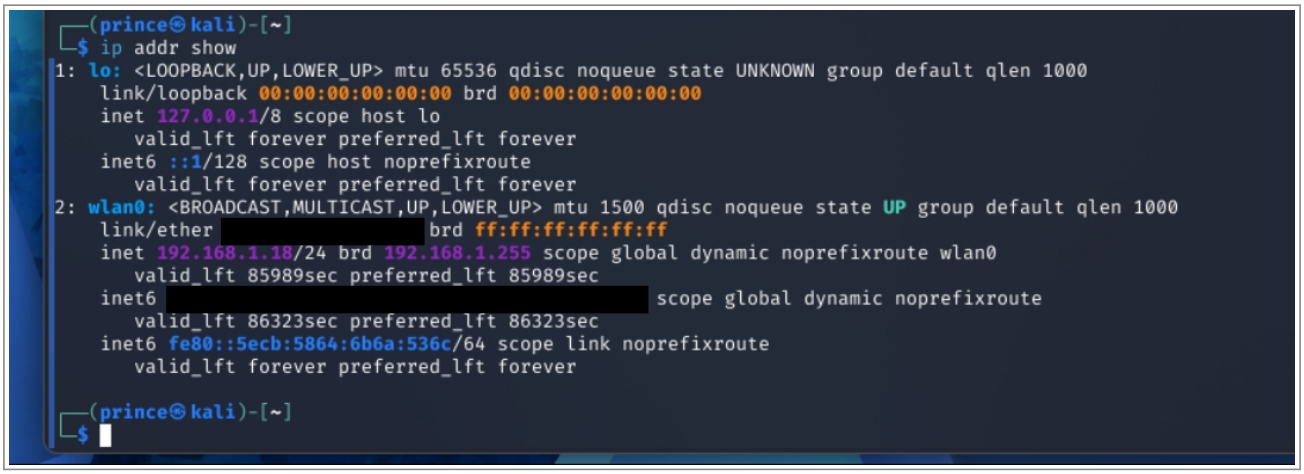


Figure 6: Output of ip addr show

What I observed: There are two interfaces. *lo* is the loopback interface (127.0.0.1). *wlan0* is the Wi-Fi interface and it is UP, with the IPv4 address **192.168.1.18/24**. It is marked *dynamic*, which means the Wi-Fi router gave the address using DHCP. IPv6 addresses are also present (the MAC address and the public IPv6 address are hidden for privacy). The name wlan0 confirms that Kali is using the real Wi-Fi card of the laptop.

Step 7: Default gateway

```
ip route
```

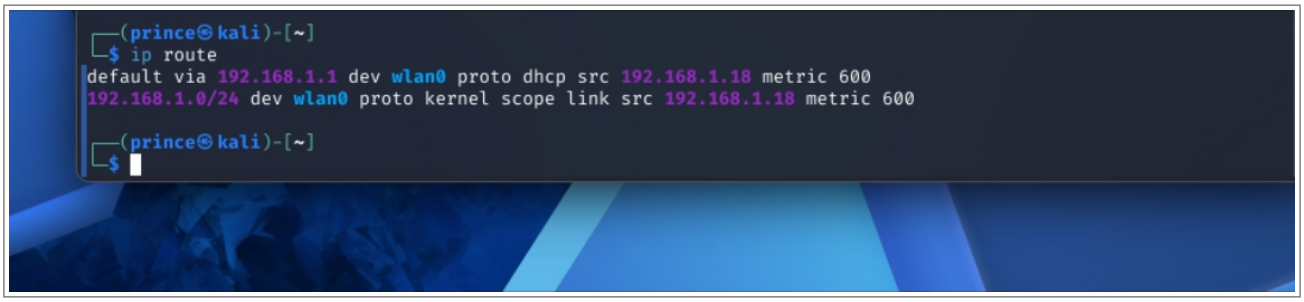


Figure 7: Output of ip route

What I observed: The default route goes through **192.168.1.1** on wlan0 and the local network is 192.168.1.0/24. All traffic to the Internet leaves through this router.

Step 8: Ping to 10.0.0.1

```
ping -c 4 10.0.0.1
```

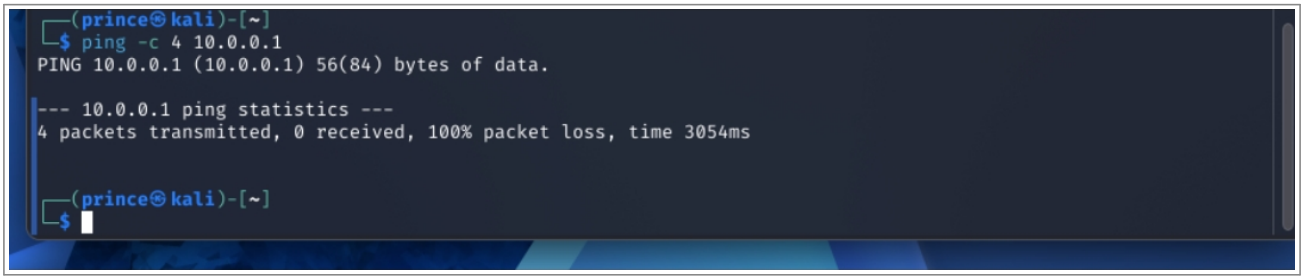


Figure 8: Output of ping -c 4 10.0.0.1

What I observed: 4 packets were sent and none came back (**100% packet loss**). 10.0.0.1 is the gateway of the VirtualBox lab network in the reference file. My laptop is on 192.168.1.0/24, so that address cannot be reached from here. This result is expected.

Step 9: Internet connection

```
ping -c 4 8.8.8.8
```

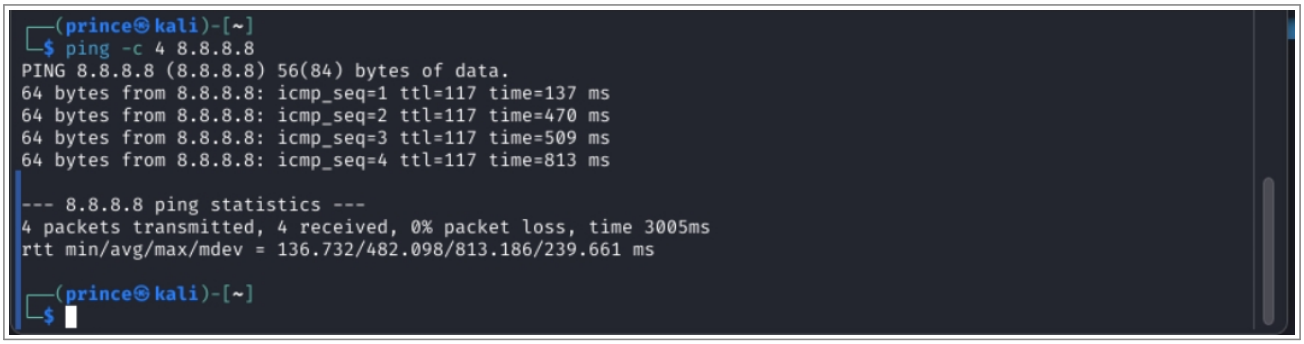


Figure 9: Output of ping -c 4 8.8.8.8

What I observed: All 4 packets came back with **0% packet loss**, so Kali has Internet access. The reply times were 137, 470, 509 and 813 ms, with an average of about 482 ms. The time changes a lot, so the connection was slow and not very steady at that moment, but it works.

7. Findings

Check	Result	Status
Kali Linux runs on the laptop	Kali GNU/Linux Rolling 2026.3 on a laptop	Pass
Runs on real hardware	Laptop chassis, real firmware, wlan0, NVMe SSD partitions	Confirmed
User and hostname	prince and kali	Pass
Memory	7.4 GiB total, 2.9 GiB available, swap unused	Pass
Storage	96.4 GB root partition on a 476.9 GB SSD	Pass
Network address	192.168.1.18/24 from DHCP	Pass
Internet	0% packet loss to 8.8.8.8	Pass
Ping to 10.0.0.1	100% packet loss (different network)	Expected

8. Difference from the Reference Lab

The reference lab (WK1-PM1) uses Kali inside VirtualBox, with a NATNetwork 10.0.0.0/24 and the IP address 10.0.0.2/24. My setup is different: Kali runs directly on the laptop, so the address comes from my Wi-Fi router and the 10.0.0.x network does not exist here.

Point	Reference lab	My setup
How Kali runs	Virtual machine in VirtualBox	Directly on the laptop
Network	NATNetwork 10.0.0.0/24	Wi-Fi network 192.168.1.0/24
Kali IP	10.0.0.2/24 (manual)	192.168.1.18/24 (DHCP)
Network card	Virtual wired adapter	Real Wi-Fi card (wlan0)

The benefit of my setup is that Kali uses the full speed of the laptop and its real Wi-Fi card. The benefit of the VirtualBox method is that the lab is fully separate and several machines can share one private network. VirtualBox can still be installed later on this laptop if a multi-machine lab is needed.

9. Conclusion and Learning

The check shows that Kali Linux is working well on my laptop. It has its own 96.4 GB partition on the SSD, about 7.4 GiB of RAM, a working Wi-Fi connection and a clear Internet connection. The ping to 10.0.0.1 failed only because that address belongs to a different network, which I now understand from the IP address and routing output.

What I learned

- How to read a machine's hardware, memory and disk layout from the terminal.
- How to tell a real machine from a virtual one using `hostnamectl` and the interface name.
- How to read `ip addr` and `ip route` to find the IP address and gateway.
- Why a ping fails when the target is in a different network, and what packet loss and reply time mean.

References

- Task material: WK1-PM1 - Lab Setup (VirtualBox and Kali Linux), Networkwalks Academy.
- Kali Linux: https://www.kali.org